Compatibility obstructions for constructive stable theta models

Independent seeds and ferromagnetic quartic networks

Drafted for Edward Baker

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Abstract

We examine explicit real-zero constructions as possible approximants to the complete theta heat flow. A normalized ferromagnetic comparison shows that any quartic network approximating theta must lose every persistent diffuse quartic site at its readout scale, even when the number of sites and all other couplings vary without bounds. Independently scaled pure-quartic convolutions, with arbitrary finite counts and an additional Gaussian component, cannot approximate theta even weakly: field growth forces infinitesimal weights, after which the limit must be Gaussian. We also collect finite interval obstructions for independent Bernoulli products, single quartic laws, and a fixed Bessel base with nonnegative even multipliers. Classical Lee–Yang theory supplies explicit coupled-quartic seeds and their Ising realizations; a certified negative sixth potential coefficient shows why a local source exclusion does not extend to every such marginal. No theta approximation theorem or conclusion about the Riemann hypothesis is obtained. Infinitesimal interacting arrays and degenerating elementary measures remain outside the exclusions.

Draft status. This manuscript was requested before formal round review. It integrates round-one Notes 01–03 and their small replay records. Derivations, certificates and authoring checks used `gpt-6.1-sol` (Codex), reasoning effort `ultra`. The checks are internal and same-model; cross-model round review and specialist human review remain pending. No literature priority is claimed.

1 Target, conventions and scope

Use the complete theta source and heat transform

$$\Phi(u)=\sum_{n\ge 1}(2\pi^2n^4e^{9u}-3\pi n^2e^{5u})e^{-\pi n^2e^{4u}},\qquad u\ge 0, \tag{1.1}$$

$$H_t(z)=\int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,\mathrm{d}u,\qquad H_0(z)=\tfrac{1}{8}\xi((1+iz)/2). \tag{1.2}$$

The source extends evenly to the real line. The xi identity and the heat normalization are classical; see Polymath [1]. In particular $\partial_t H_t=-\partial_z^2 H_t$. Set

$$I_{2k}=\int_0^\infty u^{2k}\Phi(u)\,\mathrm{d}u,\qquad m_{2k}=I_{2k}/I_0,\qquad M_\theta(h)=H_0(-ih)/I_0=\frac{\xi((1+h)/2)}{\xi(1/2)}.$$

Thus M_θ is the normalized Laplace transform of the probability density $\Phi(u)/(2I_0)$ on $\mathbb{R}$. All finite complex-field and heat derivatives converge absolutely on compact sets.

The inherited outward moment enclosure [10] gives

$$0.01155234249<m_2<0.01155265063,\qquad 2.79098616<\frac{m_4}{m_2^2}<2.79121956. \tag{1.3}$$

The strict consequence $\kappa_4(M_\theta) = m_4 - 3m_2^2 < 0$ supplies an optional check below; the new exclusion theorems have analytic proofs without this enclosure. These are source-moment enclosures, not zero certificates.

The other growth input is elementary and unconditional.

Lemma 1.1 (Theta field growth)**.** *As $h \to +\infty$,*

$$\log M_\theta(h) = \frac{h}{4}\log h - \frac{h}{4}(1 + \log(4\pi)) + \frac{7}{4}\log h + O(1). \tag{1.4}$$

In particular $\log M_\theta(h) = o(h^{4/3})$.

Proof. Insert $s = (1+h)/2$ into $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$. Stirling's formula [7, 5.11.1], with $s/2 = (h+1)/4$, and the Dirichlet-series estimate $\zeta(s) = 1 + O(2^{-s})$ give the expansion. The $\log h$ coefficient is $2 - 1/4 = 7/4$. □

We use the term stable for the real-zero Fourier or imaginary-zero Laplace property of the specified entire transform. Simplicity is not included. A construction of stable approximants converging to H_0 on every complex compact would imply RH by Hurwitz's theorem. Our results are scoped obstructions to particular constructions, rather than evidence from finite checks for that global objective.

2 Ferromagnetic comparison and persistent quartic sites

Consider a finite probability law

$$\mathrm{d}\nu(\mathbf{x}) = Z^{-1}\exp\left\{-\sum_{j=1}^{d}\alpha_j x_j^4 - \sum_j \beta_j x_j^2 + \sum_{i<j} K_{ij}x_i x_j\right\}\mathrm{d}\mathbf{x}, \quad \alpha_j > 0, \quad K_{ij} \geq 0, \tag{2.1}$$

with arbitrary real β_j, and readout $S = \sum_j b_j x_j$, $b_j > 0$. Its normalized transform is $M(h) = \mathbb{E}_\nu e^{hS}$. Quartic tails justify every finite derivative used in this section.

Absorb the readout into the coordinates:

$$Y_j = b_j x_j, \qquad a_j = \alpha_j/b_j^4, \quad c_j = \beta_j/b_j^2, \quad L_{ij} = K_{ij}/(b_i b_j). \tag{2.2}$$

Write

$$m_{a,c}(h) = \frac{\int_{\mathbb{R}} e^{-ay^4 - cy^2 + hy}\,\mathrm{d}y}{\int_{\mathbb{R}} e^{-ay^4 - cy^2}\,\mathrm{d}y}, \qquad a > 0, \quad c \in \mathbb{R}. \tag{2.3}$$

Lemma 2.1 (Normalized ferromagnetic comparison)**.** *For every real $h \geq 0$, a network (2.1) satisfies*

$$M(h) \geq \prod_{j=1}^{d} m_{a_j,c_j}(h) \geq m_{a_k,c_k}(h) \qquad (1 \leq k \leq d). \tag{2.4}$$

There is no bound on d, the other parameters, or the coupling sizes.

Proof. We give the classical GKS duplication argument in the precise continuous class used here. For any two spin monomials Y_A, Y_B, including repeated indices, the covariance is nonnegative under nonnegative fields and pair couplings. Indeed, take two copies y, z and put $u = (y+z)/2$, $v = (y-z)/2$. Twice the covariance is the expectation of $(y_A - z_A)(y_B - z_B)$ under the product

law. Each difference has nonnegative polynomial coefficients in u, v: exactly the terms with an odd total number of v's survive. The doubled interaction and field are

$$2\sum_{i<j} L_{ij}(u_i u_j + v_i v_j) + 2\sum_j h_j u_j.$$

Their exponential has nonnegative coefficients. Keep the doubled single-site density unexpanded. It is invariant independently under each $u_j \mapsto -u_j$ and $v_j \mapsto -v_j$, by evenness and exchange of the two copies. Each integrated monomial is therefore zero or nonnegative. Quartic tails justify absolute integration of the expanded interaction and polynomial factors. This proves the covariance inequality.

Let $\langle\cdot\rangle_h$ denote the common-field Gibbs law. Then

$$\frac{\mathrm{d}}{\mathrm{d}h}\langle Y_i Y_j\rangle_h = \sum_k \mathrm{Cov}_h(Y_i Y_j, Y_k) \geq 0.$$

For the normalized transform,

$$\partial_{L_{ij}} \log M(h) = \langle Y_i Y_j\rangle_h - \langle Y_i Y_j\rangle_0 \geq 0.$$

Increase the couplings from zero to L. This yields the product comparison. Each even single-site transform is an expectation of a cosh and is at least one. The proof concerns normalized transforms; dropping couplings from an unnormalized partition function would not establish the assertion. □

Lemma 2.2 (Single-site monotonicity and growth)**.** *For $h \geq 0$, $m_{a,c}(h)$ is nonincreasing in a and c. For fixed $A > 0, B \geq 0$,*

$$\log m_{A,B}(h) = \frac{3}{4^{4/3}} A^{-1/3} h^{4/3} + O(h^{2/3} + \log h). \tag{2.5}$$

An explicit lower bound, valid for $h \geq 1$, is

$$\log m_{A,B}(h) \geq F_{A,B}(h), \tag{2.6}$$
$$F_{A,B}(h) = hu - A(u + h^{-1})^4 - B(u + h^{-1})^2 - \log h - \log \frac{\Gamma(1/4)}{2A^{1/4}}, \quad u = (h/(4A))^{1/3}.$$

Proof. Differentiating the normalized expectation of $\cosh(hY)$ gives

$$\partial_a m_{a,c}(h) = -\mathrm{Cov}(\cosh(hY), Y^4), \qquad \partial_c m_{a,c}(h) = -\mathrm{Cov}(\cosh(hY), Y^2).$$

Both covariances are nonnegative: each pair consists of increasing functions of Y^2, as is seen from the independent-copy covariance identity. For (2.6), integrate the numerator over $[u, u + h^{-1}]$, and bound the denominator by $\int_{\mathbb{R}} e^{-Ay^4}\,\mathrm{d}y = \Gamma(1/4)/(2A^{1/4})$. The leading term is the positive term in (2.5). The matching upper asymptotic follows by scaling $y = h^{1/3}v$ and maximizing $v - Av^4$; the quadratic term contributes $O(h^{2/3})$. □

Theorem 2.3 (No persistent diffuse quartic site)**.** *Let M_n be normalized transforms of any sequence of networks (2.1), with dimensions and couplings allowed to vary without bounds. If $M_n(h) \to M_\theta(h)$ for every real $h \geq 0$, then for every $A > 0, B \geq 0$, eventually no site satisfies both*

$$\alpha_{n,j}/b_{n,j}^4 \leq A, \qquad \beta_{n,j}/b_{n,j}^2 \leq B. \tag{2.7}$$

In particular a fixed nontrivial quartic site cannot be retained while other ferromagnetic sites are appended to approximate theta.

Proof. If such a site persists along a subsequence, Lemmas 2.1 and 2.2 give $M_n(h) \geq m_{A,B}(h)$ along it, for every $h \geq 0$. Equations (1.4) and (2.6) supply one fixed finite h_* with $m_{A,B}(h_*) > M_\theta(h_*)$, contradicting convergence at that field. □

Corollary 2.4 (Infinitesimal readouts on bounded site shapes)**.** *Put* $\ell_{n,j} = b_{n,j}\alpha_{n,j}^{-1/4}$ *and* $r_{n,j} = \beta_{n,j}\alpha_{n,j}^{-1/2}$*. Under the hypotheses of Theorem 2.3, for every fixed real* R*,*

$$\max_{j: r_{n,j} \leq R} \ell_{n,j} \longrightarrow 0, \tag{2.8}$$

with the empty maximum defined as zero.

Proof. If $r \leq R$ and $\ell \geq \varepsilon$, then $a = \ell^{-4} \leq \varepsilon^{-4}$ and $c = r/\ell^2 \leq \max(R, 0)/\varepsilon^2$, contradicting (2.7). □

Remark 2.5 (Finite-field witness and limitations). Whenever a chosen $h_* \geq 1$ satisfies $F_{A,B}(h_*) > \log M_\theta(h_*)$, every network retaining a site in (2.7) has

$$|M(h_*) - M_\theta(h_*)| \geq e^{F_{A,B}(h_*)} - M_\theta(h_*) > 0.$$

No numerical h_* is claimed. The theorem needs convergence of exponential observables. Without an additional uniform-integrability argument, weak convergence does not permit this theorem to be applied. It does not exclude infinitesimal interacting arrays or discrete-site degenerations. Quartic field growth must not be ascribed to every Ising scaling limit; for example, critical planar limits have different tail exponents [8].

3 Independent pure-quartic convolutions

Let X have normalized density proportional to e^{-x^4}, and set $m(h) = \mathbb{E}e^{hX}$. Consider all finite products

$$M_n(h) = e^{\sigma_n^2 h^2/2} \prod_{j=1}^{d_n} m(b_{n,j}h), \qquad \sigma_n^2 \geq 0, \quad b_{n,j} \geq 0. \tag{3.1}$$

The counts and coefficients are unrestricted; the maximum scale in an empty row is defined to be zero. These represent independent scaled pure-quartic blocks plus an independent centered Gaussian. Each construction is classically stable [2, 3].

Theorem 3.1. *The transforms* (3.1) *cannot converge to* M_θ *uniformly on every complex compact. This conclusion has an entirely analytic proof and does not require a numerical theta moment enclosure.*

Proof. If $\max_j b_{n,j} \geq \varepsilon > 0$ along a subsequence, evenness gives $M_n(h) \geq m(\varepsilon h)$ for every real $h \geq 0$. Quartic growth (2.5) contradicts (1.4) at one sufficiently large fixed field. Hence $\max_j b_{n,j} \to 0$.

Derivative convergence at zero bounds the variances. With $v = \mathbb{E}X^2 > 0$,

$$\sigma_n^2 + v \sum_j b_{n,j}^2 = \operatorname{Var}(S_n) \leq C.$$

For fixed real h, the analytic expansion at zero is $\log m(bh) = vb^2h^2/2 + O_h(b^4)$, uniformly once the maximum scale is small. Since $\sum_j b_{n,j}^4 \leq (\max_j b_{n,j}^2) \sum_j b_{n,j}^2 \to 0$, summation gives

$$\log M_n(h) = \tfrac{1}{2} \operatorname{Var}(S_n)h^2 + o(1) \longrightarrow \tfrac{1}{2}v_\theta h^2, \qquad v_\theta = M_\theta''(0) > 0.$$

Thus $M_\theta(h) = e^{v_\theta h^2/2}$, contrary to (1.4). Alternatively, independence gives

$$\kappa_4(S_n) = \kappa_4(X) \sum_j b_{n,j}^4 \longrightarrow 0,$$

contradicting the inherited strict fourth-cumulant sign in (1.3). □

Corollary 3.2 (Weak convergence is also excluded)**.** *The laws represented by* (3.1) *cannot converge weakly to the normalized theta source.*

Proof. Put $v = \mathbb{E}X^2$. Integration by parts gives $\mathbb{E}X^4 = 1/4$, $\mathbb{E}X^6 = 3v/4$. Strict Cauchy–Schwarz gives $1/16 < 3v^2/4$, so $\kappa_4(X) = 1/4 - 3v^2 < 0$. Independence therefore gives $\mathbb{E}S_n^4 \le 3\,\mathrm{Var}(S_n)^2$. For positive variances, Paley–Zygmund applied to S_n^2 shows $\mathbb{P}\{S_n^2 > \mathrm{Var}(S_n)/2\} \ge 1/12$. Tightness forces bounded variances.

The function $\log m(h)/h^2$ extends continuously at zero, is bounded on finite intervals and tends to zero at infinity by (2.5). Hence $m(h) \le e^{Ch^2}$ for all real h. The variance bound supplies a common sub-Gaussian bound for all M_n. It gives uniform integrability of every fixed exponential observable. Weak convergence would thus give $M_n(h) \to M_\theta(h)$ for real h. On any complex disc, $|M_n(z)| \le M_n(|z|)$, so normality and the identity theorem upgrade this to complex-compact convergence, contradicting Theorem 3.1. □

The potential cone in Section 4 is not the reason for this exclusion. Indeed, for independent X, Y of the pure-quartic law, the source of $X + Y/4$ has a positive fourth and a negative sixth potential coefficient. A finite exact calculation makes this explicit. For general $\delta > 0$, put $r = \delta^4$, $A = 1 + r$, and $v = \Gamma(3/4)/\Gamma(1/4)$. Expanding $\int e^{-(s-\delta y)^4 - y^4}\,\mathrm{d}y$ at $s = 0$ gives

$$[s^4](-\log \rho_\delta) = \frac{1 - (11/2)r + r^2 + 18rv^2}{A^3},$$
$$[s^6](-\log \rho_\delta) = \frac{8\delta^2 v}{A^{9/2}}\big(-(1-r)^2 + 9rv^2\big).$$

Differentiation is justified by quartic decay. The finite expansion uses $\mathbb{E}y^{4k} = (1/4)_k/A^k$ and $\mathbb{E}y^{4k+2} = (v/\sqrt{A})(3/4)_k/A^k$. For $\delta = 1/4$, strict Cauchy bounds $1/12 < v^2 < 1/4$ and exact rational arithmetic [11] give

$$0.97 < [s^4](-\log \rho_{1/4}) < 0.99, \qquad -0.25 < [s^6](-\log \rho_{1/4}) < -0.11.$$

Thus leaving the local potential cone can occur even without inter-block interaction; Theorem 3.1 supplies the stronger global approximation obstruction.

The result concerns fixed pure-quartic elementary shapes. It does not exclude variable quadratic tilts, dependent blocks, or a sequence with changing elementary distributions outside this class. Its proof uses field growth and cumulants, not an assumed location of theta zeros.

4 Finite obstructions for other seed classes

4.1 Independent Bernoulli products

For real b_j, every zero r of a finite product $\prod_j \cos(b_j z)$ forces a zero at qr, for every odd integer q. The same holds after multiplication by a zero-free entire function. The property is closed under nonzero complex-compact limits: Rouché's theorem gives approximant zeros $r_n \to r$, and compact convergence at qr_n then forces a limiting zero at qr.

The complete-field Acb certificate [11] gives, with

$$a = 28.26945, \quad b = 28.26946, \quad B = [3a, 3b],$$

Input	Outward rational interval for H_0
a	$(2 \cdot 10^{-11}, 3 \cdot 10^{-11})$
b	$(-9 \cdot 10^{-10}, -8 \cdot 10^{-10})$
Every real $x \in B$	$(-3.68 \cdot 10^{-13}, -3.67 \cdot 10^{-13})$

Thus H_0 has a zero $r \in (a, b)$ and is nonzero at $3r$, without any assumption about simplicity or the first zero.

Proposition 4.1. *No independent Bernoulli cosine products, even with arbitrary counts and zero-free entire multipliers, converge complex-compactly to H_0. Moreover, on $K = [a, b] \cup B$, every real continuous function $f(x) = g(x) \prod_j \cos(b_j x)$ with $g \neq 0$ on K satisfies*

$$\|f - H_0\|_{C(K)} \geq 3.6 \cdot 10^{-13}. \tag{4.1}$$

Proof. The qualitative assertion follows from odd dilation and the certificate. For the quantitative assertion, an error strictly below the stated constant would preserve the opposite signs at a, b. The intermediate value theorem supplies a zero $r_f \in (a, b)$. A cosine factor must vanish there, and hence at $3r_f$, contradicting the certified margin on B. □

4.2 A source-potential inequality

For a positive source ρ, write $W(x) = -\log \rho(\sqrt{x})$, and let Δ_h denote a forward difference in x. If

$$-\log \rho(u) = c + \beta u^2 + \alpha u^4 + \sum_{k \geq 3} v_k u^{2k}, \qquad v_k \geq 0, \tag{4.2}$$

with convergence at the tested nodes, then $\Delta_h^3 W(0) \geq 0$. Terms of degree at most two in x cancel, and higher monomials have positive third differences. Normalization and the quadratic heat weight do not change the test. The complete-source certificate gives

$$\Delta^3_{1/100}[-\log \Phi(\sqrt{x})]_{x=0} \in (-0.000434999, -0.000434997). \tag{4.3}$$

Consequently sources in (4.2) cannot converge to theta even at these four source nodes. We do not infer a general weak-limit exclusion for a growing-degree cone from this pointwise test alone.

Proposition 4.2 (Single quartic laws)**.** *Normalized laws proportional to $e^{-\alpha_n u^4 - \beta_n u^2}$, $\alpha_n > 0, \beta_n \in \mathbb{R}$, cannot converge weakly to the normalized theta source.*

Proof. If $r_n = \|(\alpha_n, \beta_n)\| \to \infty$, pass to a subsequence of normalized coefficient vectors with limit (a, b) of norm one. The polynomial $p(u) = au^4 + bu^2$ is nonconstant. Choose bounded intervals A, B with $\inf_A p > \sup_B p + 2\eta$. Uniform convergence of the normalized potentials on these intervals gives

$$\mu_n(A) \leq (|A|/|B|) e^{-r_n \eta} \mu_n(B) \longrightarrow 0,$$

contradicting positive theta mass on A. Thus the parameters are bounded. Extract their limits. The normalizer is determined by the mass of any compact interval of positive limiting mass, so local density convergence identifies the limit as a quartic or Gaussian law. Its third difference is zero, contrary to (4.3). Integrability rules out a nonintegrable boundary potential. Normalized Fourier convergence would imply weak convergence by Lévy's continuity theorem and is excluded as well. □

4.3 Fixed Bessel base with positive even coefficients

Let

$$B_0(u) = e^{-2\pi \cosh 4u}, \qquad \int_0^\infty B_0(u)\cos(zu)\,\mathrm{d}u = \tfrac{1}{4}K_{iz/4}(2\pi). \tag{4.4}$$

The Bessel seed has only real zeros; see [6, 7]. Even universal multipliers U, with $U(iz) \in \mathcal{LP}$ and $U \geq 0$ on $\mathbb{R}$, preserve this property [5, Proposition 1]. Their canonical products have the form

$$U(u) = Cu^{2m}e^{au^2}\prod_j (1 + u^2/r_j^2), \qquad C > 0, \quad a \geq 0.$$

In particular all even Taylor coefficients are nonnegative. The following larger-class exclusion does not require universality.

Proposition 4.3. *Suppose $U_n(u) = \sum_{k\geq 0} c_{n,k}u^{2k}$ are nonzero even entire functions with $c_{n,k} \geq 0$, and B_0U_n are integrable nonnegative sources. Their even source measures cannot converge weakly to the theta source, nor can their cosine transforms converge to H_0 on every complex compact.*

Proof. Every $U_n(\sqrt{x})$ has nonnegative fourth forward differences. The complete-source certificate instead gives

$$\Delta^4_{1/1000}[\Phi(\sqrt{x})e^{2\pi\cosh(4\sqrt{x})}]_{x=0} \in (-5.232\cdot 10^{-7}, -5.230\cdot 10^{-7}). \tag{4.5}$$

For the weak-limit assertion, bounded masses imply, for every $R \geq 0$,

$$U_n(R)\int_R^{R+1} B_0(u)\,\mathrm{d}u \leq \int_0^\infty B_0(u)U_n(u)\,\mathrm{d}u \leq C.$$

Since $|U_n(z)| \leq U_n(|z|)$, the multipliers form an entire normal family. A compactly convergent subsequence gives local uniform density convergence to B_0U, which weak convergence identifies with Φ. This contradicts (4.5). Fourier convergence implies bounded masses and, after normalization, weak convergence by Lévy's theorem. □

De Bruijn already discusses the universal-factor approximation obstruction for this fixed Bessel base [4, p. 221]. Equation (4.5) is a finite diagnostic of the stated coefficient class, not a priority claim for that classical obstruction. Varying or degenerating Bessel bases and multipliers outside the coefficient cone remain outside Proposition 4.3.

5 Admissible coupled seeds and their precise limitations

5.1 Classical stability and exact heat evolution

For (2.1), the continuous-spin Lee–Yang theorem gives nonvanishing of the Laplace transform when all site fields have positive real part [2, Theorems 1 and 3]; see also [3, Theorem 1]. Every site is an even quartic law, the couplings are nonnegative, and positive readout weights put all site fields in the same half-plane. Spin-flip symmetry gives the negative half-plane as well. The transform $M(iz)$ therefore has only real zeros. The same fixed law, weighted by e^{tS^2}, has

$$\beta_j(t) = \beta_j - tb_j^2, \qquad K_{ij}(t) = K_{ij} + 2tb_ib_j, \qquad \partial_t Z = \partial_h^2 Z, \tag{5.1}$$

where $Z(t,h) = \mathbb{E}_\nu e^{tS^2+hS}$ retains its initial normalizer. All finite complex derivatives converge on compact sets by quartic domination. The real-zero conclusion is asserted for real $t \geq 0$.

5.2 An explicit seed outside the potential cone

Take

$$\mathrm{d}\nu(x,y) = Ce^{-x^4-y^4+(7/2)(x^2+y^2)+xy}\,\mathrm{d}x\,\mathrm{d}y, \qquad S = x+y. \tag{5.2}$$

It is initially interacting. Its heat parameters are $\beta(t) = -7/2 - t$ and $K(t) = 1 + 2t$. For $s = x + y$, $v = (x - y)/2$, its marginal is proportional to

$$e^{-s^4/8+2s^2}\int_{\mathbb{R}} e^{-2v^4+6v^2-3s^2v^2}\,\mathrm{d}v.$$

Let $W = V^2$ under the normalized law $V \propto e^{-2(V^2-3/2)^2}$. Then

$$[s^6](-\log\rho_S(s)) = \tfrac{9}{2}\kappa_3(W). \tag{5.3}$$

The outward integral certificate [11] encloses this coefficient by

$$-0.022766643413912614837902926830890898 42 \quad \text{with radius } 7.60\cdot 10^{-39}.$$

Thus this marginal is outside (4.2). It invalidates an extension of that local cone exclusion to all ferromagnetic marginals. It is nevertheless not a theta source: Theorem 2.3 excludes this fixed seed, and also appending sites while retaining one of its fixed diffuse readout sites.

5.3 Elementary Ising realization, with a paid convergence domain

The construction is a specialization of the classical quartic Ising approximation [2]. For two even-size blocks of N unit spins, let M_1, M_2 be their magnetizations and set

$$c_N = (12N^3)^{-1/4}, \qquad J_{\text{within}} = 1/N + 7c_N^2, \qquad J_{\text{between}} = c_N^2, \qquad S_N = c_N(M_1+M_2). \tag{5.4}$$

Each J is the coefficient of an unordered spin pair. Every coupling is positive at time zero.

We spell out the convergence payment for this exact normalization. For even N and $M = 2r$, the centered binomial ratio is

$$R_N(M) = \frac{\binom{N}{(N+M)/2}}{\binom{N}{N/2}} = \prod_{j=1}^{|r|}\frac{1-(2j-1)/(N+1)}{1+(2j-1)/(N+1)}.$$

The inequality $\log[(1-u)/(1+u)] \le -2u - 2u^3/3$ gives

$$\log R_N(M) \le -\frac{M^2}{2(N+1)} - \frac{M^2(M^2-2)}{12(N+1)^3}. \tag{5.5}$$

Writing $x = c_N M$, the within-block weight therefore has the uniform central-ratio envelope

$$R_N(M)e^{J_{\text{within}}M^2/2} \le e^{-x^4/8+(7/2+\sqrt{12})x^2}. \tag{5.6}$$

On fixed x-compacts, expansion of the same exact ratio gives

$$\log R_N(M) = -\frac{M^2}{2N} - \frac{M^4}{12N^3} + o(1);$$

the error is uniform there, since $M = O(N^{3/4})$. Thus the central ratio tends to $e^{-x^4+(7/2)x^2}$. The cross-block factor is e^{xy}.

Use associated piecewise-constant histograms of mesh $2c_N$. Inequality (5.6), with slightly weaker constants inside each shrinking cell, gives an integrable quartic envelope. The cross term is absorbed by $|xy| \le (x^2+y^2)/2$. The local limits and dominated convergence give convergence of normalizers and densities to (5.2). The actual lattice readout differs from the histogram readout by at most the shrinking mesh. On each compact complex (t,h) set the same quartic envelope absorbs $e^{t(x+y)^2+h(x+y)}$ and each fixed derivative power. The normalized heat transforms and every fixed derivative consequently converge complex-compactly to the specified continuous seed.

At finite N, heat adds $2tc_N^2$ to every pair and retains the diagonal scalar $e^{2tNc_N^2}$, with the initial normalizer fixed. This gives (5.1) exactly before taking the limit. This argument is for a fixed two-block architecture and its explicit continuous limit; it gives no theta approximation or estimate uniform in an increasing number of blocks.

6 Certificates, controls and research disposition

The replay records contain small outward enclosures rather than large generated data. Their source hashes identify the implementations and do not replace the arithmetic or the analytic implications above. The dependencies used were Python 3.10.0, python-flint 0.9.0 and FLINT 3.6.0; the arithmetic contracts are documented in [9].

Record / evaluator	Fixed retained run and scope
Harmonic sign certificate	160 bits, three complete xi evaluations, a 30-second cap. One ball covers the entire tripled interval; no source cutoff.
Source alternative screens	160 bits, nine source evaluations with eight theta terms and a proved complete tail; 30-second cap and required difference width below 10^{-7}.
Two-quartic coefficient	160 bits, four Acb integrals on $[0,4]$; goals 2^{-130}, at most 100,000 evaluations and depth 100 per integral.
Unequal quartic convolution	Exact rational enumeration through degree six at $\delta = 1/4$; three coefficients, no quadrature or parameter grid.

For the source screens, positivity and geometric summation give

$$0 < T_N(u) \le \frac{2\pi^2 e^{9u}(N+1)^4 e^{-\pi(N+1)^2}}{1-16e^{-\pi(2N+3)}} \qquad (u \ge 0).$$

For the four integrals defining W, the omitted tails satisfy, for $u \ge 4$, $0 \le k \le 3$,

$$u^{2k}e^{-2(u^2-3/2)^2} \le e^{-u^4/2} \le e^{-32u}, \qquad \int_4^\infty e^{-32u}\,\mathrm{d}u = e^{-128}/32.$$

Both intervals and quadrature contracts are therefore paid independently of agreement between floating approximations. Scouts preceded the retained certificate contracts and are identified as such in the records. The new field-growth results in Sections 2 and 3 require no additional numerical campaign.

The repository's collision controls [12] screen any proposed general theta implication. A claim of stable complex-compact approximation to an Euler-factor control with $0 < c < 1$ fails at time zero, where its explicit extra factor already has nonreal zeros. For the weighted-square, reflected, and smooth/positive-primitive positive-threshold controls, the corresponding convergence fails on a positive pre-collision slice with a nonreal zero. Hurwitz gives the contradiction on that slice. A double real zero at the collision time itself is compatible with Lee–Yang. We have not assigned an uncomputed coefficient or parameter representation to any control.

The results narrow the constructive search in concrete ways. Independent Bernoulli products and independent scaled pure-quartic blocks cannot give the desired limit, even with changing counts. Single quartic laws and the specified fixed Bessel multiplier class are also excluded. Interactions do change zero geometry, and coupled quartic marginals can leave the simple source-potential cone. Nevertheless, any successful quartic network must satisfy the degeneration condition (2.7); appending sites while retaining a fixed diffuse quartic seed cannot suffice.

What remains is a construction in the infinitesimal or degenerating interacting regime, with an actual theta approximation error tending to zero and uniform heat/differentiation bounds. Neither the classical stability theorem, the explicit seed, nor the exclusion results supply that construction. The investigation remains open in that stated regime, pending formal review of the present round.

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